PSFormer: Pyramid Superpixel Transformer for Hyperspectral Image Classification

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***Abstract*— Hyperspectral image (HSI) classification is a core processing procedure in the remote sensing community, which has been recently well studied using vision transformers (ViTs). However, due to the high computational and memory complexities, existing transformer-based classification methods tend to restrict the spatial extent of the transformer to small cropped HSI patches instead of the whole HSI data, thus sacrificing the essential strength of transformers in long-range interaction modeling and overlooking the beneficial multiscale features in HSI data. Inspiringly, here we propose PSFormer, a novel pyramid superpixel transformer (PSFormer) method specifically for HSI classification, in order to make full use of the transformer to excavate multiscale local–global features in HSI data. Specifically, a progressive superpixel merging strategy is introduced to flexibly control the scale of feature maps. Furthermore, a unique transformer backbone design based on a spectral attention layer and a classification head with a gate mechanism are developed, to adaptively exploit valuable local–global information at different scales with low computational cost. Extensive experimental results on five widely used datasets demonstrate the superiority of PSFormer over other state-of-the-art networks. For the sake of reproducibility, the related code of the PSFormer method will be open-sourced at: https://github.com/immortal13.**

***Index Terms*— Hyperspectral image (HSI) classification, local–global, superpixel segmentation, transformer.**

I. Introduction

HYPERSPECTRAL image (HSI) consists of dozens to hundreds of narrow and contiguous spectral bands, covering a wide range of wavelengths in the same scene. The abundant spectral and spatial information in HSI empowers it to distinguish visually similar materials [1]. Therefore, it has been an extremely valuable data source for numerous applications, including precision agriculture, military reconnaissance, and medical diagnosis [2], [3], [4]. HSI classification is a basic task and core procedure for these applications, aiming to identify the category of every pixel.

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Deep learning (DL) techniques, particularly convolutional neural networks (CNNs), have become a preferable choice compared with conventional approaches in HSI classification tasks, due to the capability of effectively preserving the spatial structure and hierarchically encoding the local information of HSI [5], [6]. However, the CNN-based networks typically overlook the long-range dependencies beyond small cropped HSI patches and cannot flexibly adapt to different HSI scenarios at inference using convolutional kernels with fixed shape and static weights [7]. Meanwhile, graph convolutional networks (GCNs) have gained significant attention in HSI classification, which generalizes convolutional operations to graphs with flexible structure and can learn relatively long-range contextual information [8]. To address the challenge of processing a large number of HSI pixels, numerous GCN-based networks aggregated pixels of HSI data into superpixel nodes and performed superpixelwise feature extraction [9], where each superpixel represented a contiguous group of pixels with perceptual meanings [10], [11]. These approaches can effectively reduce the number of graph nodes and facilitate the delineation of irregular object boundaries. Nevertheless, existing superpixel-based GCNs still face limitations regarding receptive fields as they primarily focus on information propagation between adjacent superpixels. In general, as shown in Fig. 1(a), the restricted receptive field of convolutional operations remains a bottleneck for effectively extracting long-range contextual information in both CNNs and GCNs [12].

Recently, Dosovitskiy et al. [13] pioneered employing the ViT to replace the convolutional paradigm for image classification, which decomposed an image into small patches (i.e., tokens) and captured their long-range dependencies with the dynamic self-attention (SA) mechanism. Despite the distinct capability of long-range feature interaction and adaptive information aggregation, ViT is a columnar framework that generates single-scale and low-resolution feature maps, making it difficult to process multiscale variation and preserve fine-grained details [14]. To alleviate this issue, emerging methods incorporated the feature pyramid structure into ViT [15], [16], [17], which demonstrates great performance for dense prediction tasks that require multiscale information and image details. Although transformer models can effectively model long-range pixel relationships when applied to typical vision tasks, it is infeasible to directly adapt it to the HSI classification task. Mainly because the computational cost of the attention layer in the transformer grows quadratically with

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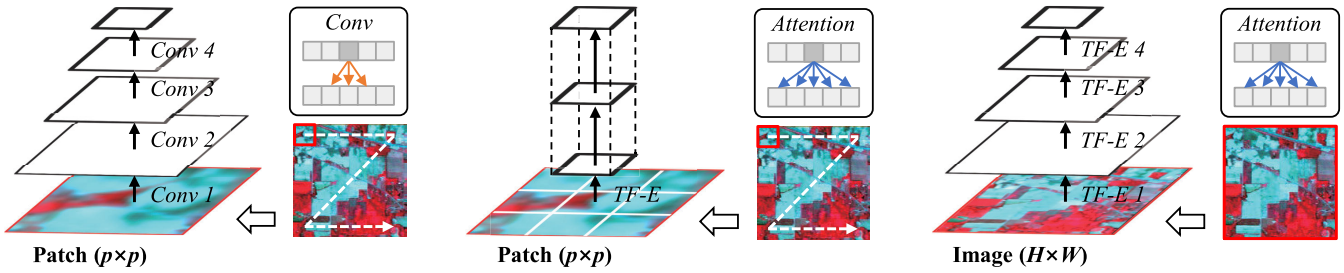


Fig. 1. Comparisons of different HSI classification architectures, where "Conv" and "TF-E" denote "convolution" and "transformer encoder," respectively. (a) CNN-based and GCN-based classification architectures. (b) Transformer-based classification architecture. (c) PSFormer classification architecture. Unlike other architectures that are limited to leveraging information within small cropped HSI patches or operate exclusively at a single spatial scale, PSFormer can control the scale of features and leverage multiscale local–global dependencies from the entire HSI data for classification.

the number of tokens [18], prohibiting its application to large HSI datasets (e.g., the Houston University (HU) dataset with 349 × 1905 pixels). More recently, as depicted in Fig. 1(b), several transformer-based classification methods have been developed by applying transformers with columnar structure on the cropped HSI patch of size $p \times p$ (e.g., 11 × 11) around each pixel [19], [20], in order to reduce the computational consumption. Despite the efforts of existing transformer-based HSI classification approaches, there are two major problems that need to be further explored. On the one hand, because of the high computational complexity of transformer models, existing methods tend to follow the conventional patch-based learning framework, which imposes a strong prior assumption that neighboring pixels belong to the same class and limits the spatial extent of transformers to the cropped HSI patch rather than the whole HSI. While this framework is particularly beneficial for HSI classification tasks with limited training samples, it compromises the key advantage of transformers in modeling long-range relationships and results in slow inference speeds. On the other hand, the output features of existing methods are generally on a single spatial scale, making it hard to explore the beneficial contextual information of objects residing in different spatial scales and leading to feature redundancy.

In view of these problems, we propose a novel pyramid superpixel transformer (PSFormer) for HSI classification, which is not tied to the conventional patch-based learning framework and benefits from multiscale spatial contextual information, as illustrated in Fig. 1(c). The PSFormer is composed of a stem module, a PSFormer backbone, and a classification head with a gate mechanism. Specifically, we first adopt a stem module to transform the raw HSI into low-level features with fixed channels, in order to unify feature dimension and minimize image redundancy. After that, the low-level features are fed into a PSFormer backbone with L_1 stages to enhance nonlocal interactions at different scales, where each stage contains a patch embedding layer and L_2 transformer encoder layers. We exploit the progressive superpixel merging strategy to flexibly control the scale transformation of features between every two stages with low computational costs. At last, a simple classification head with a gate mechanism is devised to adaptively integrate multiscale feature representations for classification. In general, the goal of PSFormer is to extract beneficial multiscale spatial contextual information from the entire HSI by adopting the progressive superpixel merging strategy and making several key designs to the transformer model, in order to make full use of transformer for superior classification accuracy.

The main contributions of this work are as follows.

1) We develop a PSFormer for HSI classification, which can effectively model multiscale local–global dependencies of the whole HSI without slicing it into small patches, thereby accentuating strengths and avoiding weaknesses of the ViT when adapting it to the HSI classification tasks.
2) We introduce a progressive superpixel merging strategy, along with a PSFormer backbone and a classification head with gate mechanism. These modules work together to flexibly control feature scales at a low computational cost and adaptively exploit multiscale local–global features.
3) Extensive experimental results on three benchmark HSI datasets and two unmanned aerial vehicle (UAV) HSI datasets demonstrate that PSFormer outperforms other state-of-the-art networks.

The rest of this article is organized as follows. Section II revisits the related work. Section III describes the details of the proposed PSFormer. Section IV exhibits extensive experimental results and comprehensive analyses. Section V draws the conclusion of this article and provides limitations and plausible future work.

II. Related Works

A. Traditional Methods

Numerous attempts have been made to achieve high HSI classification performance. Machine learning (ML) classifiers such as support vector machine (SVM) were first introduced for HSI classification. After that, some dimension reduction methods were applied prior to classification to extract the intrinsic spectral information, which include the principal component analysis (PCA), geodesic-based sparse manifold hypergraph (GSMH) [21], and locality preserving projections (LPPs) [22]. However, the classification results of these spectral-based methods are heavily affected by the spectral variability in HSIs. Therefore, a variety of spatial features, such as extended morphological profile (EMP) [23], were

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developed and integrated with spectral features to further improve classification accuracy. In addition, superpixel algorithms have been leveraged as a preprocessing step of HSI classification to incorporate spatial information, which cluster pixels that share similar properties and can well preserve object boundaries [24]. For instance, an unsupervised superpixelwise PCA (SuperPCA) method was proposed to capture the intrinsic low-dimensional features of HSIs for classification [25]. In [26], a superpixel graph learning approach was introduced, which used superpixels generated by an improved simple linear iterative clustering (SLIC) algorithm as nodes for a traditional graph classifier, in order to smooth classification maps while retaining sharp edges. Nevertheless, these classification methods heavily depend on prior information from experts and fail to extract the underlying nonlinear information in HSI.

B. CNN-Based and GCN-Based Methods

Due to the ability to effectively retain spatial structure and hierarchically extract local features of HSI, CNNs have become the mainstream for HSI classification tasks. For instance, Li et al. [27] devised a two-stream 2-D CNN architecture for HSI classification, which separately captured spectral information and spatial information. In order to fully exploit the structural attributes of 3-D HSI cubes, Paoletti et al. [28] developed a 3-D CNN model for the joint extraction of spectral–spatial features. In addition, residual connections, dense connections, and feedback connections were introduced for developing deeper or wider CNNs [29], which can further promote classification accuracy without incurring model degradation. On the contrary, several lightweight CNN methods [30] have been designed to achieve competitive performance with limited training samples and computing resources. Furthermore, the attention mechanism was embedded into CNN frameworks to explore contextual dependencies in the spectral or spatial domain [31]. While numerous CNN-based classification networks have been proposed, most of them still encounter challenges in effectively leveraging information beyond short-range Euclidean neighbors.

In contrast, GCNs offer the advantage of processing non-Euclidean data structure and capturing relatively long-range dependencies [8]. To mitigate the computational cost incurred by large numbers of pixels in HSI, numerous GCN-based HSI classification methods treated HSI as a superpixel graph, where each node represented a superpixel and edges connected spatially adjacent superpixels. Liu et al. [32] constructed multilevel superpixel graphs by hierarchically merging adjacent superpixels to exploit multiscale features of HSI data. Furthermore, Zhang et al. [7] introduced a differentiable superpixel segmentation algorithm that refines superpixel boundaries during network training to construct a superpixel graph, and devised a novel mixhop superpixel-based GCN to extract local features within superpixels and long-range features between adjacent superpixels for HSI classification. Nevertheless, these methods tend to propagate information between adjacent superpixels, thereby limiting direct connections to more distant ones. Several hybrid methods [8], [33] that combine CNN and GCN models have also been explored, however, effectively capturing long-range features in HSI data using convolutional operations with small receptive fields remains nontrivial.

C. Transformer-Based Methods

ViTs have been a disruptive trend in various computer vision (CV) tasks [34]. Compared with CNN and GCN architectures, the purely attention-based transformer models can flexibly adapt to the input HSI data and have the natural tendency to capture long-range spatial dependencies in HSIs. Thus, several transformer-based methods have been developed for HSI classification tasks [35], [36], [37]. For instance, Hong et al. [19] developed a transformer-based method to capture spectral contextual dependencies in HSI, which can adapt to pixel-wise and patchwise inputs. In [38], a local transformer was proposed to extract detailed spatial features, which can reduce misclassification in the boundary. In order to exploit both the spectral and spatial features, a two-stream spectral–spatial transformer was designed in [39] to separately capture the dependencies between spectral sequences and spatial sequences of HSI patches. Moreover, Qiao et al. [40] devised a transformer architecture to capture multiscale neighborhood features using local attention with different receptive fields. Inspired by the MetaFormer [41], Li et al. [42] introduced a lightweight self-pooling transformer using two parameter-free modules as spectral and spatial token mixers, in order to achieve superior performance while significantly reducing the number of model parameters. Nevertheless, in order to decrease the computational loads, these methods often resort to the patch-based learning framework and apply the transformer model to cropped HSI patches, defying the main motivation of modeling long-range dependencies with the transformer.

To address this limitation, several image-based global learning methods have emerged by incorporating superpixel segmentation techniques into the transformer, which leverage the model's implicit receptive field rather than explicit patch size, thereby enhancing the utilization of nonlocal information in HSI data. For example, Shi et al. [43] transformed the HSI into a single-scale superpixel graph and designed a graph-guided transformer to capture global and local correlations among superpixel nodes. Yu et al. [44] integrated a CNN block and a ViT encoder to cooperatively exploit local and nonlocal information in HSI, where superpixels with single fixed scale were adopted as input tokens for the ViT branch to reduce computational cost. However, the output feature maps of these methods suffer from low resolution and are confined to a single spatial scale, resulting in contextual information loss and blurry details. Unlike these approaches that are limited to leveraging information within small cropped HSI patches or operate exclusively at a single spatial scale and generate low-resolution feature maps, we propose a transformer-based classification method that can flexibly control the scale of features and adaptively integrate multiscale local–global dependencies from the entire HSI data, aiming to accentuate the strengths of ViT while mitigating their weaknesses when adapting it to HSI classification tasks.

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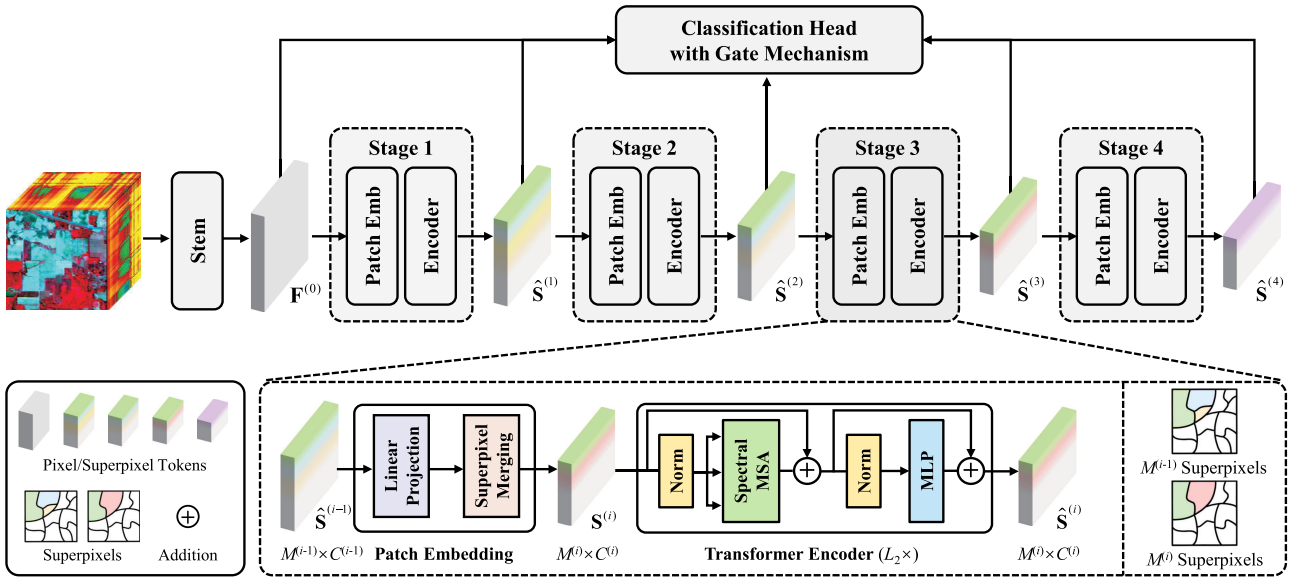


Fig. 2. Overall framework of PSFormer. The PSFormer framework consists of three modules: stem module, PSFormer backbone, and classification head with gate mechanism. Note that four stages are used for illustration, and the number of stages in PSFormer is not limited to four.

D. Overall Architecture

The overall framework of PSFormer is presented in Fig. 2. The first part of PSFormer is a stem module, which transforms the whole HSI data into low-level features to reduce image redundancy and unify feature dimensions. The convolutional stem consists of a batch normalization (BN) layer, a 1 $\times$ 1 convolutional layer, and a leaky rectified linear unit (LeakyReLU) activation function. Then, we devise a PSFormer backbone with L_1 stages, where each stage comprises a patch embedding layer followed by L_2 transformer encoder layers. The progressive superpixel merging strategy is adopted to transform the scale of feature maps inside the transformer backbone. More specifically, the patch embedding layer starts with a linear projection layer that projects input features to an arbitrary dimension, followed by a superpixel merging module to implement scale transformation of features. The output features are fed into stacked transformer encoder layers to further enhance the discriminative ability. Take the first stage of a four-stage transformer backbone for example, the output pixel features $\mathbf{P} \in \mathbb{R}^{H\times W\times C}$ of the stem module are flattened to $\mathbf{F}^{(0)} \in \mathbb{R}^{N\times C}$ and passed through the linear projection layer to generate $\mathbf{S}^{(0)} \in \mathbb{R}^{N\times C^{(1)}}$, and then projected into the superpixel features $\mathbf{S}^{(1)} \in \mathbb{R}^{M^{(1)}\times C^{(1)}}$ with the superpixel merging operation, where H, W, and C denote the height, width, and dimension of hidden features, respectively, $N = H \times W$ denotes the total pixel number, $C^{(1)}$ is the channel dimension at the first stage, and $M^{(1)}$ is the superpixel number at the first stage. Subsequently, $\mathbf{S}^{(1)} \in \mathbb{R}^{M^{(1)}\times C^{(1)}}$ are fed into a transformer encoder with L_2 layers and obtain enhanced superpixel features $\hat{\mathbf{S}}^{(1)} \in \mathbb{R}^{M^{(1)}\times C^{(1)}}$. Similarly, with the superpixel features from the previous stage as input, the following superpixel features of a four-stage transformer backbone can be derived: $\hat{\mathbf{S}}^{(2)}$, $\hat{\mathbf{S}}^{(3)}$, and $\hat{\mathbf{S}}^{(4)}$. At last, we design a classification head with gate mechanism that first projects superpixel features $\hat{\mathbf{S}}^{(i)} \in \mathbb{R}^{M^{(i)}\times C^{(i)}}$ to flattened pixel features $\mathbf{F}^{(i)} \in \mathbb{R}^{N\times C}$, and then adaptively integrates multiscale information for classification. The cross-entropy loss is employed to optimize the parameters in PSFormer. The details of the progressive superpixel merging strategy, PSFormer backbone, and classification head with gate mechanism are elaborated in Section III.

III. PSFormer

A. Progressive Superpixel Merging Strategy

Unlike existing transformer backbone networks [13], [14], [15], [16] that maintain the same resolution throughout the network or reduce the spatial dimension using patch merging/progressive shrinking strategy, the proposed PSFormer method introduces a progressive superpixel merging strategy, which embeds the superpixel hierarchy into the transformer network with differentiable matrix multiplication operations and merges neighboring superpixel features in each stage to decrease feature dimensions, as illustrated in Fig. 3. In this manner, we can flexibly construct a superpixel feature pyramid with low computation cost together with suppressing redundant information and preserving edge details.

Specifically, the original HSI data $\mathbf{H} \in \mathbb{R}^{H\times W\times B}$ is processed into noncorrelated features with three bands, where B denotes the band number. After that, we perform superpixel segmentation with the super hierarchy (SH) algorithm [11] to generate a fine-to-coarse hierarchy $\{\mathcal{S}^{(1)}, \mathcal{S}^{(2)}, \mathcal{S}^{(3)}, \mathcal{S}^{(4)}\}$ with superpixel numbers of $M^{(i)}$ $(i = \{1, 2, 3, 4\})$, note that four levels are used for illustration purposes. We add an extra $\mathcal{S}^{(0)}$ with N pixels (i.e., $M^{(0)}$ pixels) to bridge the pixel-level feature space and superpixel-level feature space. Different from other superpixel segmentation methods that produce a fixed number of superpixels at a time, the SH method can yield any amount of superpixels concurrently by computing a minimum spanning tree (MST) with Boruvka's algorithm. In order to integrate the superpixel hierarchy into the transformer backbone and avoid the nondifferentiability

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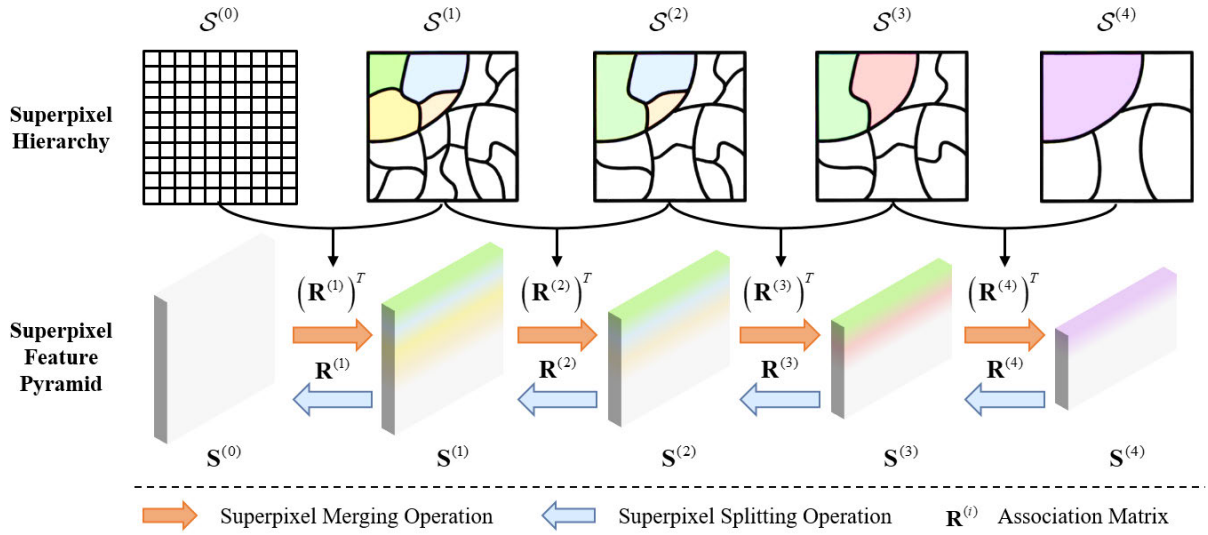


Fig. 3. Illustration of the progressive superpixel merging strategy. Note that four levels are used for illustration, and the level of superpixel hierarchy is not limited to four.

issue, we compute the association matrix $\mathbf{R}^{(i)} \in \mathbb{R}^{M^{(i-1)} \times M^{(i)}}$ between superpixels at different scales [32], [45] as follows:

$$\mathbf{R}_{p,q}^{(i)} = \begin{cases} 1, & \text{if } \mathcal{S}_p^{(i-1)} \subseteq \mathcal{S}_q^{(i)} \\ 0, & \text{otherwise} \end{cases} \tag{1}$$

where $\mathcal{S}_p^{(i-1)}$ is the pth superpixel in the superpixel map $\mathcal{S}^{(i-1)}$, and $\mathcal{S}_q^{(i)}$ is the qth superpixel in the superpixel map $\mathcal{S}^{(i)}$. Finally, the superpixel merging operation and the inverse superpixel splitting operation can be implemented by differentiable matrix multiplications, that is,

$$\begin{aligned} \mathbf{S}^{(i)} &= \text{Merge}\left(\mathbf{S}^{(i-1)}\right) = \left(\hat{\mathbf{R}}^{(i)}\right)^{\mathrm{T}} \mathbf{S}^{(i-1)} \\ \mathbf{S}^{(i-1)} &= \text{Split}\left(\mathbf{S}^{(i)}\right) = \tilde{\mathbf{R}}^{(i)} \mathbf{S}^{(i)} \end{aligned} \tag{2}$$

where $\mathbf{S}^{(i-1)}$ and $\mathbf{S}^{(i)}$ represent the superpixel features in the $(i-1)$th stage and ith stage of the transformer backbone, $\hat{\mathbf{R}}^{(i)}$ denotes the column-normalized $\mathbf{R}^{(i)}$, and $\tilde{\mathbf{R}}^{(i)}$ denotes the row-normalized $\mathbf{R}^{(i)}$.

B. PSFormer Backbone

The PSFormer backbone comprises L_1 stages that generate pyramid features. Each stage shares a similar architecture, which consists of a patch embedding layer and L_2 transformer encoder layers. More specifically, the patch embedding layer includes a linear projection layer and a superpixel merging module. The former performs channel dimension transformation, note that we adopt a BN layer and an LeakyReLU activation function before and after the layer. The latter implements scale transformation with the aid of association matrices, in order to learn rich contextual representations and reduce feature redundancy simultaneously.

Each transformer encoder layer is composed of an attention layer and a feed-forward layer, where the attention layer is a key component effective in extracting long-range information. However, the conventional attention layer has quadratic complexity to the number of tokens (i.e., image patches), prohibiting its application to large HSI datasets. Therefore,

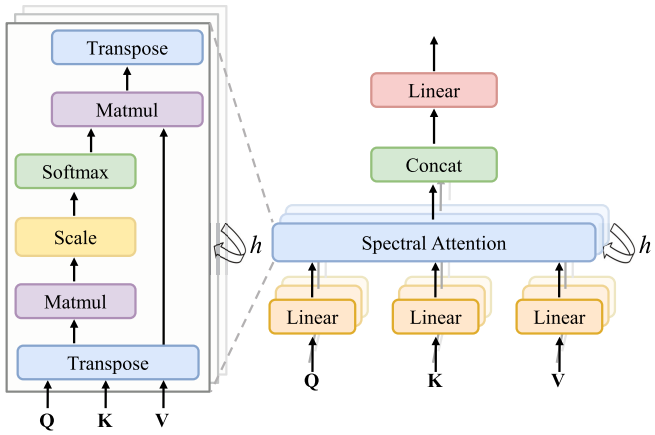


Fig. 4. Illustration of the spectral MSA layer.

we propose the spectral SA layer with linear complexity (see Fig. 4), which models global dependencies through computing attention values across spectral dimension rather than spatial dimension. Concretely, for the spectral SA layer in stage i, the input features $\mathbf{S}^{(i)} \in \mathbb{R}^{M^{(i)} \times C^{(i)}}$ are first linearly projected into the queries $\mathbf{Q} \in \mathbb{R}^{M^{(i)} \times C^{(i)}}$, keys $\mathbf{K} \in \mathbb{R}^{M^{(i)} \times C^{(i)}}$, and values $\mathbf{V} \in \mathbb{R}^{M^{(i)} \times C^{(i)}}$. Subsequently, $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$ are transposed before the attention operation to obtain transposed matrices $\mathbf{Q}' \in \mathbb{R}^{C^{(i)} \times M^{(i)}}$, $\mathbf{K}' \in \mathbb{R}^{C^{(i)} \times M^{(i)}}$, and $\mathbf{V}' \in \mathbb{R}^{C^{(i)} \times M^{(i)}}$. The enhanced values are then calculated as follows:

$$\begin{aligned} \mathbf{A} &= \text{softmax}\left(\frac{\mathbf{Q}'\left(\mathbf{K}'\right)^{\mathrm{T}}}{\sqrt{M^{(i)}}}\right) \\ \hat{\mathbf{V}} &= \mathbf{A}\mathbf{V}' \end{aligned} \tag{3}$$

where $\mathbf{A} \in \mathbb{R}^{C^{(i)} \times C^{(i)}}$ is the attention score matrix, $\hat{\mathbf{V}} \in \mathbb{R}^{C^{(i)} \times M^{(i)}}$ denotes the matrix of enhanced values, and $\sqrt{M^{(i)}}$ is the scaling parameter. The enhanced representations $\hat{\mathbf{S}}^{(i)} \in \mathbb{R}^{M^{(i)} \times C^{(i)}}$ can be obtained by transposing the output values $\hat{\mathbf{V}}$. For the spectral multihead SA (MSA) layer, we divide queries,

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TABLE I
NUMBER OF TRAINING AND TESTING SAMPLES FOR EACH CLASS OF THREE HSI DATASETS

No.	Salinas (SA) dataset				Pavia University (PU) dataset				Houston University (HU) dataset			
	Color	Class name	Training	Test	Color	Class name	Training	Test	Color	Class name	Training	Test
1		Brocoli green weeds 1	30	1979		Asphalt	30	6601		Grass healthy	198	1053
2		Brocoli green weeds 2	30	3696		Meadows	30	18619		Grass stressed	190	1064
3		Fallow	30	1946		Gravel	30	2069		Grass synthetic	192	505
4		Fallow rough plow	30	1364		Trees	30	3034		Tree	188	1056
5		Fallow smooth	30	2648		Painted metal sheets	30	1315		Soil	186	1056
6		Stubble	30	3929		Bare soil	30	4999		Water	182	143
7		Celery	30	3549		Bitumen	30	1300		Residential	196	1072
8		Grapes untrained	30	11241		Self-Blocking bricks	30	3652		Commercial	191	1053
9		Soil vinyard develop	30	6137		Shadows	30	917		Road	193	1059
10		Corn senesced green weeds	30	3248						Highway	191	1036
11		Lettuce romaine 4wk	30	1038						Railway	181	1054
12		Lettuce romaine 5wk	30	1897						Parking lot1	192	1041
13		Lettuce romaine 6wk	30	886						Parking lot2	184	285
14		Lettuce romaine 7wk	30	1040						Tennis court	181	247
15		Vinyard untrained	30	7238						Running track	187	473
16		Vinyard vertical trellis	30	1777								

keys, and values into several heads (i.e., four heads) and perform SA operations in parallel.

C. Classification Head

The classification head initially maps superpixel features $\hat{\mathbf{S}}^{(i)} \in \mathbb{R}^{M^{(i)} \times C^{(i)}}$ into flattened pixel features $\mathbf{F}^{(i)} \in \mathbb{R}^{N \times C}$ using differentiable matrix multiplications and 3×3 blueprint separable convolution (BSConv) blocks [7], [46]. The former implements scale transformation, while the latter performs channel dimension mapping and enhances local details. Subsequently, it integrates multiscale information for classification.

In order to effectively integrate all pyramid features for HSI classification, we present several designs for the classification head. The "Add Head" elementwise sums up the flattened pixel features $\mathbf{F}^{(i)}$ $(i = \{0, 1, 2, 3, 4\})$, and feds the output features into a softmax classifier that contains a linear projection layer and a softmax activation function. Similar to the "Add Head," the "Concat Head" concatenates pixel features of all stages and uses the softmax classifier to perform classification. Considering that features of the early stages are high-resolution and rich in low-level spatial details, while features of the deeper stages have insufficient resolution and contain more high-level semantic information, the simple fusion strategies including addition and concatenation may encounter difficulties in filling the gaps in resolution and level [47]. Thus a classification head with a gate mechanism (i.e., "Gate Mechanism Head") is devised to obtain the importance coefficients of pyramid features and adaptively integrate features from all scales.

Concretely, we first compute the importance coefficients $(\boldsymbol{\alpha}^{(0)}, \boldsymbol{\alpha}^{(1)}, \boldsymbol{\alpha}^{(2)}, \boldsymbol{\alpha}^{(3)}, \boldsymbol{\alpha}^{(4)})$ as follows:

$$\left(\boldsymbol{\alpha}^{(0)}, \boldsymbol{\alpha}^{(1)}, \boldsymbol{\alpha}^{(2)}, \boldsymbol{\alpha}^{(3)}, \boldsymbol{\alpha}^{(4)}\right) = \mathrm{g}\left(\mathbf{F}^{(0)}, \mathbf{F}^{(1)}, \mathbf{F}^{(2)}, \mathbf{F}^{(3)}, \mathbf{F}^{(4)}\right) \quad (4)$$

where $g(\cdot)$ represents the gate mechanism and $\boldsymbol{\alpha}^{(0)}$, $\boldsymbol{\alpha}^{(1)}$, $\boldsymbol{\alpha}^{(2)}$, $\boldsymbol{\alpha}^{(3)}$, and $\boldsymbol{\alpha}^{(4)} \in \mathbb{R}^{N \times 1}$ indicate the corresponding importance coefficients of $\mathbf{F}^{(0)}$, $\mathbf{F}^{(1)}$, $\mathbf{F}^{(2)}$, $\mathbf{F}^{(3)}$, and $\mathbf{F}^{(4)}$ for all pixels. For the nth pixel, the importance coefficients can be calculated by

$$\begin{aligned} \omega_n^{(i)} &= \mathrm{sigmoid}\left(\mathbf{w}_n \cdot \left(\mathbf{f}_n^{(i)}\right)^{\mathrm{T}}\right) \\ \alpha_n^{(i)} &= \mathrm{softmax}\left(\omega_n^{(i)}\right) = \frac{\exp\left(\omega_n^{(i)}\right)}{\sum_{k=0}^{4} \exp\left(\omega_n^{(k)}\right)} \end{aligned} \quad (5)$$

where $\mathbf{w}_n \in \mathbb{R}^{1 \times C}$ represents a shared weight vector for all pyramid features of the nth pixel, $\mathbf{f}_n^{(i)} \in \mathbb{R}^{1 \times C}$ is the feature of the nth pixel at stage i, and $\omega_n^{(i)}$ and $\alpha_n^{(i)}$ denote the importance coefficient and the normalized coefficient of the nth pixel at stage i, respectively. Then, the final output feature $\mathbf{F}$ can be obtained by

$$\mathbf{F} = \sum_{k=0}^{4} \left(\boldsymbol{\alpha}^{(k)} \cdot \mathbf{F}^{(k)} + \mathbf{F}^{(k)}\right). \quad (6)$$

Note that the residual connection is used to enhance the stability. At last, the output features are fed into the softmax classifier to predict the class probabilities of all pixels in HSI.

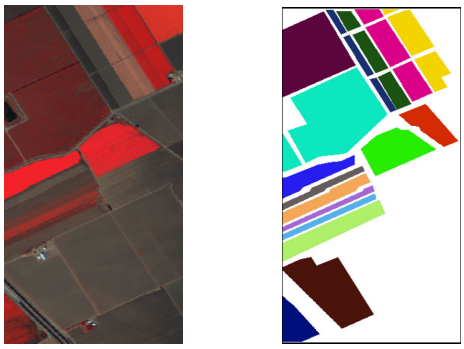
IV. EXPERIMENTAL RESULTS AND ANALYSIS

A. Dataset Description

Three benchmark HSI datasets are used to evaluate the performance of PSFormer, including Salinas (SA), Pavia University (PU), and HU. In the following, we illustrate each dataset in detail.

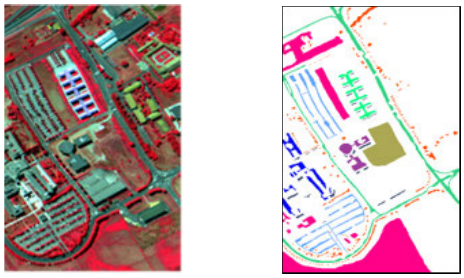
1) *SA:* The airborne visible/infrared imaging spectrometer (AVIRIS) sensor collected data over the SA Valley, California at 3.7-m pixels. The image comprises 512×217 pixels and 204 spectral channels. 16 classes representing different land cover types are defined in the dataset, such as "Grapes," "Lettuce," and "Vinyard".

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(a) (b)

Fig. 5. SA dataset. (a) False color image. (b) Ground-truth map.



(a) (b)

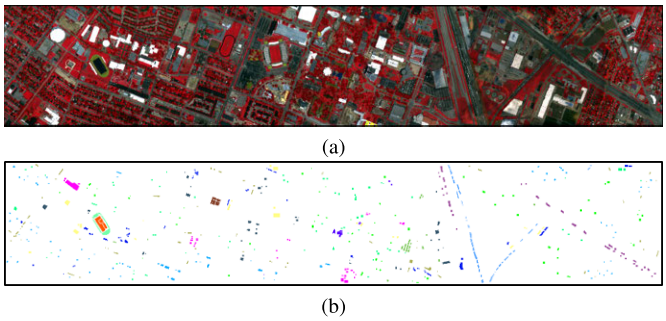
Fig. 6. PU dataset. (a) False color image. (b) Ground-truth map.

The complete category information and the number of training and test pixels for each category are listed in Table I. The false color image and ground-truth map of the SA dataset are displayed in Fig. 5.

2) *PU:* The PU dataset was gathered by the reflective optics system imaging spectrometer (ROSIS) sensor over the city of Pavia, Italy, which comprises 610 × 340 pixels with 1.3-m per pixel resolution and has 103 spectral channels. The ground truth contains nine classes, including "Trees," "Bare Soil," and "Bitumen." Table I shows the amounts of training and test data in each category. Fig. 6 displays the false color image and ground-truth map.
3) *HU:* The HU dataset was provided in the 2013 Geoscience and Remote Sensing Society (GRSS) Data Fusion Contest, which was acquired by the compact airborne spectrographic imager (CASI) sensor over the University of Houston campus and its neighboring area. This image has 349 × 1905 pixels with 2.5-m per pixel resolution and 144 spectral bands, and the ground-truth map is divided into 15 categories (see Table I). Note that we use the cloud-free version to be consistent with other methods [19], [48]. Fig. 7 shows the false color image and the corresponding ground-truth map.

B. Experimental Setup

Our experiments are performed using Pytorch backend on an NVIDIA server with a Tesla V100 GPU. The numbers of epochs are set to 500, 600, and 300 for the SA, PU, and HU datasets, respectively. We employ the Adam optimizer with a learning rate equal to 0.0005 and remain it unchanged throughout the training process. In terms of the training and test sets, two sampling strategies are exploited to comprehensively validate the effectiveness of our PSFormer method. Concretely, on the SA and PU datasets, we randomly select 30 training samples for each class from the labeled pixels and use the other pixels as test data. On the HU dataset, the spatially disjoint training–testing sets provided by the 2013 IEEE GRSS Data Fusion Contest are used. In addition, we repeat each experiment five times and adopt the overall accuracy (OA), average accuracy (AA), kappa coefficient, and per-class accuracy (CA) as evaluation criteria.



(a) (b)

Fig. 7. HU dataset. (a) False color image. (b) Ground-truth map.

To evaluate the classification performance of PSFormer, we conducted a comprehensive comparison with nine HSI classification methods, including two traditional methods, i.e., SVM and SuperPCA [25] (Superpixel PCA), and seven state-of-the-art DL-based methods, i.e., homogeneous pixel detection module-spectral partition residual network (HPDM-SPRN) [49], adaptive spectral–spatial kernel ResNet (A^2S^2K-ResNet) [50], group-aware hierarchical transformer (GAHT) [51], spectral–spatial feature tokenization transformer (SSFTT) [52], GCN and transformer fusion network (GTFN) [53], dual context network with transformer (DCN-T) [54], and graph guided transformer (G^2T) [43].

For the SVM, the radial basis function (RBF) kernel is selected by default, and fivefold cross validation is adopted to search for the optimal hyperparameters γ in the range of {0.1, 0.01, 0.001} and C in the range of {1, 10, 100, 1000}. For the SuperPCA and DL-based comparison methods, we adopt the default values from open-source codes and further carefully tune the hyperparameters to achieve optimal performance. Regarding the proposed PSFormer, the channel dimensions of pyramid feature C and $C^{(i)}$ ($i = \{1, 2, 3, 4\}$) are set to 128, 64, 32, 16, and 8 referring to [32]. The Transformer encoder comprises a four-head SA layer, an MLP, two LN layers, and a Gaussian error linear unit (GELU) nonlinear activation layer. Some hyperparameters (i.e., the number of stages L_1 and encoder layers L_2, and a number of superpixels in each stage $M^{(i)}$ ($i = \{1, 2, 3, 4\}$)) in the proposed PSFormer method influence classification performance to a certain extent, and they will be analyzed in the parameter sensitivity experiments.

C. Classification Results

Tables II–IV summarize the classification results of different methods on SA, PU, and HU datasets, where the

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TABLE II
ACCURACY COMPARISONS ON THE SA DATASET WITH 30 TRAINING SAMPLES

ID	Traditional methods		State-of-the-art DL-based methods							Our method
	SVM	SuperPCA	HPDM-SPRN	A^2S^2K-ResNet	GAHT	SSFTT	GTFN	DCN-T	G^2T	PSFormer
1	97.41±1.92	99.98±0.03	**100±0**	**100±0**	99.93±0.09	99.95±0.11	99.97±0.07	99.52±0.6	**100±0**	**100±0**
2	98.82±0.62	99.72±0.13	**100±0**	99.98±0.03	99.43±0.69	99.88±0.28	99.95±0.08	99.1±1.64	99.96±0.08	99.84±0.32
3	97.43±1.48	99.57±0.16	**100±0**	99.74±0.13	95.66±2.64	**100±0**	99.93±0.1	99.19±1.62	99.98±0.04	**100±0**
4	99.5±0.15	99.69±0.11	99.55±0.7	97.24±1.51	97.46±3.65	99.93±0.07	99.81±0.15	96.96±1.96	99.88±0.14	**100±0**
5	95.77±2.38	98.45±0.08	96.85±1.52	**99.66±0.35**	95.81±2.69	97.95±1.7	97.54±2.73	97.86±1.51	98.64±0.51	98.56±0.41
6	98.98±0.39	98.15±0.81	99.41±0.71	**100±0**	99.15±1.7	99.96±0.06	99.64±0.39	98.17±1.93	99.71±0.27	99.84±0.17
7	99.43±0.11	98.23±1.49	99.97±0.03	**100±0**	99.5±0.26	99.95±0.11	99.93±0.08	98.7±1.63	99.66±0.17	99.97±0.02
8	71.16±7.84	**98.39±1.65**	91.58±4.82	92.07±3.33	84.59±9.68	90.41±3.35	88.22±3.75	92.67±0.48	94.04±2.59	97.22±2.39
9	97.33±1.29	92.57±6.48	**100±0**	99.66±0.18	99.74±0.5	99.92±0.14	99.92±0.11	97.37±2.54	**100±0**	**100±0**
10	88.12±5.6	96.79±1.05	98.73±0.86	98.67±0.44	97.16±1.99	98.03±1.23	96.75±2.88	**99.53±0.68**	97.34±1.17	97.99±0.67
11	96.88±0.89	96.86±1.33	99.94±0.12	98.87±2.25	99.73±0.49	**99.96±0.08**	99.94±0.09	99.47±1	99.65±0.19	99.40±0.67
12	99.05±0.99	99.32±0.63	98.57±1.48	99.74±0.28	99.65±0.29	99.88±0.2	99.53±0.2	99.88±0.2	99.47±0	**100±0**
13	99.19±0.34	98.28±0.08	99.59±0.81	**100±0**	99.01±0.9	99.65±0.57	99.84±0.23	99.83±0.2	97.99±1.6	99.39±0.25
14	95.37±3.03	96.71±1.63	98.87±0.9	98.79±0.76	**99.79±0.27**	95.35±7.32	98.71±0.8	99.28±0.5	97.48±1.94	98.50±1.22
15	67.02±5.92	**99.41±0.13**	96.83±2.13	82.51±4.62	87.14±8.26	92.31±4.17	91.43±5.43	97.36±1.88	96.7±1.78	97.96±0.53
16	95.81±2.3	99.37±0.47	**100±0**	99.59±0.75	98.83±1.04	98.95±1.02	99.59±0.26	99.99±0.03	97.54±1.22	99.01±0.61
OA	87.54±1.76	97.96±0.61	97.44±0.99	95.44±0.56	94.19±1.55	96.58±0.95	95.95±0.83	97.24±0.81	97.84±0.41	**98.84±0.54**
AA	93.58±2.20	98.22±0.25	98.74±0.39	97.91±0.20	97.04±0.45	98.25±0.85	98.17±0.41	98.43±0.54	98.63±0.18	**99.23±0.24**
Kappa	86.15±1.96	97.73±0.68	97.15±1.10	94.91±0.62	93.54±1.71	96.20±1.05	95.49±0.92	96.93±0.90	97.59±0.45	**98.71±0.61**

TABLE III
ACCURACY COMPARISONS ON THE PU DATASET WITH 30 TRAINING SAMPLES

ID	Traditional methods		State-of-the-art DL-based methods							Our method
	SVM	SuperPCA	HPDM-SPRN	A^2S^2K-ResNet	GAHT	SSFTT	GTFN	DCN-T	G^2T	PSFormer
1	73.85±3.59	81.62±7.08	92.87±5.86	93.79±8.99	86.58±6.63	90.16±1.46	84.93±7.84	91.26±4.76	98.24±1.15	**98.64±1.08**
2	76.71±5.87	88.02±3.1	96.55±2.51	**98.89±0.66**	90.6±2.55	94.76±5.86	92.22±6.9	94.7±3.95	94.69±4.41	98.42±0.70
3	75.99±4.12	96.98±1.22	92.44±6.26	83.62±9.47	84.83±5.88	89.57±4.8	85.37±8.23	89.56±3.97	**97.76±1.52**	97.10±2.47
4	89.55±2.49	71.1±6.47	94.09±3.23	91.65±5.36	90.15±4.21	91.88±1.03	91.28±3.77	88.85±3.48	92.38±3.59	**96.88±1.75**
5	99.21±0.1	98.3±0.62	**100±0**	99.92±0.07	99.16±0.58	99.63±0.2	99.59±0.23	99.7±0.32	99.82±0.04	**100±0**
6	82.23±7.31	95.78±1.16	99.46±0.55	84.7±8.71	91.74±4.42	95.81±1.8	97.33±1.13	99.69±0.49	**99.86±0.14**	99.78±0.27
7	89.69±4.06	96.31±1.27	**100±0**	88.57±7.85	97.05±1.98	98.52±1.2	99.69±0.36	96.57±3.11	99.08±0.68	99.94±0.03
8	80.59±5.68	95.06±1.34	94.5±5.38	77.7±6.95	96.58±0.79	87.25±3.61	90.2±6.34	97.33±1.25	98.53±0.96	**99.90±0.03**
9	99.85±0.09	99.85±0.09	97.62±1.38	98.37±1.34	93.65±3.78	96.21±3.46	97.67±2.54	95.64±2.09	99.52±0.28	**99.96±0.05**
OA	79.72±1.78	88.60±2.01	96.00±2.00	91.94±2.20	90.84±1.02	93.08±2.80	91.69±3.47	94.54±1.40	96.56±1.80	**98.69±0.30**
AA	85.3±3.70	91.45±1.17	96.39±1.49	90.80±1.55	92.26±0.45	93.55±0.83	93.14±1.33	94.81±0.89	97.76±0.48	**98.96±0.10**
Kappa	74.10±1.98	85.16±2.53	94.74±2.59	89.47±2.81	88.03±1.29	90.93±3.55	89.17±4.33	92.83±1.76	95.49±2.34	**98.27±0.40**

highest values are marked in bold and the second highest values are underlined. It shows that SVM is generally inferior to deep networks on all datasets, and SuperPCA cannot perform well on the PU and HU datasets, due to their insufficient ability to excavate high-level features that model visual semantics. In terms of the deep networks, we observe that the CNN-based methods can obtain high accuracy with great consistency for all datasets, due to their strong capability in local information extraction. Conversely, patch-based transformer methods (i.e., GAHT, SSFTT, and GTFN) exhibit inferior and inconsistent performance. For instance, the SSFTT method performs well on the SA and PU datasets while struggling on the HU dataset. Mainly because these patch-based transformer methods discard the essential advantage of transformers in long-range context modeling and fail to utilize the beneficial multiscale information, making it difficult to retain global consistency and capture the spatial context in various HSI scenes. While DCN-T inputs the entire HSI data to extract image-level spectral–spatial features, it requires a relatively large amount of training samples to

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TABLE IV
ACCURACY COMPARISONS ON THE HU DATASET WITH DISJOINT TRAINING–TESTING SETS PROVIDED BY THE 2013 IEEE GRSS DATA FUSION CONTEST

ID	Traditional methods		State-of-the-art DL-based methods							Our method
	SVM	SuperPCA	HPDM-SPRN	A^2S^2K-ResNet	GAHT	SSFTT	GTFN	DCN-T	G^2T	PSFormer
1	83.57	83.67	78.27±0.64	**99.46±0.15**	76.58±2	90.16±1.46	84.98±5.59	58.16±1.4	82.36±0.72	82.64±0.49
2	98.21	98.21	90.28±2.88	95.87±2.79	97.24±1.77	94.76±5.86	86.3±2.94	83.48±0.28	90.32±4.59	**97.61±1.71**
3	99.8	99.8	97.27±0.15	**100±0**	99.25±0.55	94.72±3.54	99.8±0.44	55.21±0.33	99.6±0	**100±0**
4	98.3	98.39	90.34±3.32	98.23±0.8	91.88±4.06	89.93±3.94	92.22±2.88	73.88±0.16	95.95±1.49	**98.94±0.23**
5	98.77	98.49	99.66±0.32	99.38±0.88	99.05±0.84	99.74±0.28	**100±0**	80.57±0.16	99.6±0.26	**100±0**
6	97.9	99.3	95.24±0.69	98.84±0.82	89.51±5.92	97.44±2.25	95.94±0.31	**99.72±0.63**	91.89±3.44	95.80±0.00
7	87.13	88.43	72.22±2.07	**99.4±0.64**	80.58±2.75	78.39±7.06	80.65±1.61	83.62±0.5	89.78±4.67	93.69±2.88
8	71.32	57.17	87.45±4.45	69.68±4.74	78.84±5.48	62.78±8.54	**93.16±2.38**	52.63±0.79	81.01±9.23	73.90±2.62
9	66.01	82.15	83.59±2.27	71.8±3.7	85.02±1.44	89.77±2.67	86.1±3.03	90.78±0.24	**92.33±2.8**	84.23±2.67
10	**89.67**	76.35	88.05±4.36	84.34±7.23	78.63±8.36	64.77±4.8	59.96±7.46	77.66±0.89	70.04±17.06	77.39±11.54
11	84.73	87.86	**98.25±0.96**	82.53±6.31	94.88±3.39	91.52±5.04	90.66±5.12	79.35±0.35	88.9±6.01	88.33±6.91
12	61.77	68.68	91.91±1.87	**94.27±2.74**	80.73±2.11	90.36±1.96	92.24±0.99	60.94±1.06	83.05±7.8	90.55±1.76
13	62.81	65.61	72.07±2.65	90.55±11.66	88.77±2.09	**91.7±2.03**	85.9±7.62	75.79±0	84.28±3.03	64.28±3.33
14	99.19	99.6	**100±0**	97.75±3.19	**100±0**	**100±0**	**100±0**	85.26±0.54	99.92±0.16	**100±0**
15	97.25	96.83	99.92±0.17	98.72±1.11	94.16±4.71	99.22±0.8	97.89±2.05	22.58±0.28	99.58±0	**100±0**
OA	85.13	85.22	88.78±0.61	88.87±1.05	87.57±1.09	85.67±0.46	88.00±1.37	71.93±0.09	88.60±0.97	**89.41±1.14**
AA	86.43	86.70	89.63±0.42	**92.05±0.60**	89.01±1.17	88.16±0.29	89.72±0.98	71.98±0.06	89.91±0.76	89.83±1.04
Kappa	83.85	83.96	87.83±0.65	87.92±1.15	86.51±1.18	84.50±0.50	86.98±1.48	69.49±0.09	87.61±1.05	**88.50±1.23**

achieve satisfactory classification performance. In contrast, the image-based G^2T method integrates pixel-level features extracted by convolutional blocks with single-scale superpixel-level features derived from graph-guided SA modules, leading to high accuracies on three datasets.

It can be seen that the proposed PSFormer method outperforms other traditional methods and state-of-the-art networks for all three datasets. In detail, when compared with the second best method, PSFormer delivers a significant improvement by 0.88% of OA, 0.49% of AA, and 0.98% of Kappa coefficient on the SA dataset, by 2.13% of OA, 1.20% of AA, and 2.78% of Kappa coefficient on the PU dataset, and by 0.54% of OA and 0.58% of Kappa coefficient on the HU dataset. This is due to twofold reasons. For one thing, PSFormer can directly model multiscale long-range relationships under the image-based learning framework, which maximizes the inherent strength of transformers. For another, with the progressive superpixel merging strategy and key modifications to transformer model, PSFormer can adaptively integrate both low-level information to preserve details and high-level features to model semantics.

The classification maps obtained by comparison methods are displayed in Figs. 8–10. It can be observed that the classification maps of SVM have severe salt and pepper noise, and SuperPCA reduces noise by integrating spatial features into the spectral classifier. The CNN and patch-based transformer methods yield better visual performance but still exhibit misclassifications in some categories such as "Bare soil" on the PU dataset and "Tree" on the HU dataset. In contrast, the image-based transformer methods (i.e., DCN-T, G^2T, and PSFormer) demonstrate better global consistency and higher accuracy in boundary regions. However, DCN-T suffers from the over-smoothing problem; that is, the details of land covers (e.g., small "Commercial" around relatively large "Residential") are eroded. Meanwhile, G^2T misclassifies "Grapes untrained" and "Vinyard untrained" on the SA dataset because of their high spectral similarity. By integrating beneficial contextual features and capturing long-range dependencies, the classification maps of PSFormer contain less misclassification and show better consistency with ground truth.

D. Effectiveness of Different Modules in PSFormer

We conduct experiments on three HSI datasets to investigate how each module of PSFormer works in terms of accuracy and efficiency. Table V summarizes the overall accuracies on three datasets using different architectures. The "Single Scale/4" denotes a columnar framework that uses fine-grained image patches (e.g., 4 × 4 pixels per patch) as input and output single-scale feature map, the "Single Scale/32" denotes a similar structure using coarse image patches (e.g., 32 × 32 pixels per patch) as input, and the "Progressive Shrinking" adopt the pyramid structure to obtain features with strides 4, 8, 16, and 32 pixels relative to the input image. It can be observed that using the high-resolution columnar architecture (i.e., Single Scale/4) leads to unsatisfactory and unstable classification performance. This might be explained by the fact that the high-resolution columnar architecture makes it difficult to capture coarse high-level semantic features with limited training samples. The low-resolution columnar framework

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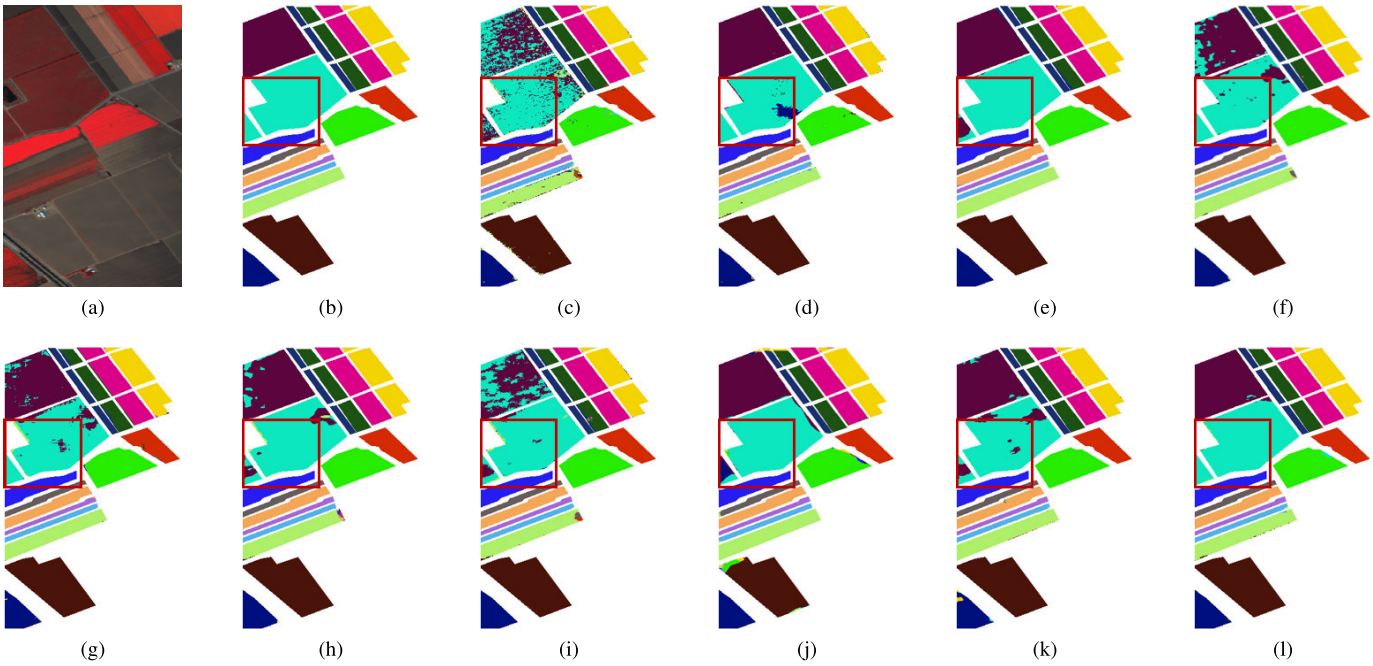


Fig. 8. Classification maps generated by different methods on the SA dataset. (a) False color image. (b) Ground-truth map. (c) SVM (OA = 87.54%). (d) SuperPCA (OA = 97.96%). (e) HPDM-SPRN (OA = 97.44%). (f) A^2S^2K-ResNet (OA = 95.44%). (g) GAHT (OA = 94.19%). (h) SSFTT (OA = 96.58%). (i) GTFN (OA = 95.95%). (j) DCN-T (OA = 97.24%). (k) G^2T (OA = 97.84%). (l) PSFormer (OA = 98.84%). We use red boxes to highlight the areas where improvements have been made.

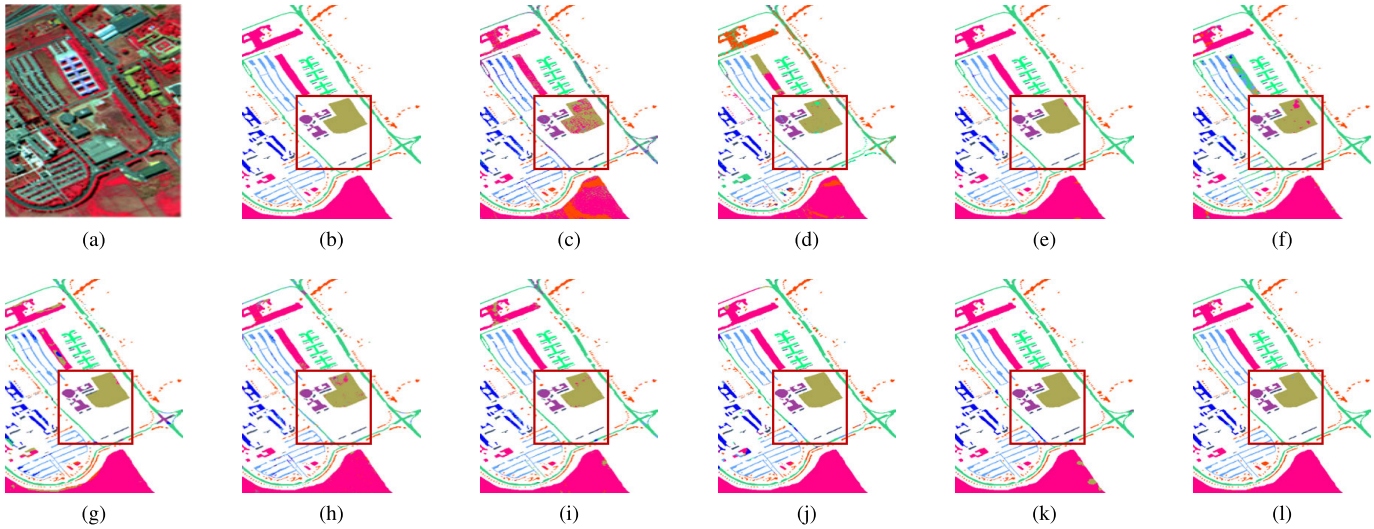


Fig. 9. Classification maps generated by different methods on the PU dataset. (a) False color image. (b) Ground-truth map. (c) SVM (OA = 79.72%). (d) SuperPCA (OA = 88.60%). (e) HPDM-SPRN (OA = 96.00%). (f) A^2S^2K-ResNet (OA = 91.94%). (g) GAHT (OA = 90.84%). (h) SSFTT (OA = 93.08%). (i) GTFN (OA = 91.69%). (j) DCN-T (OA = 94.54%). (k) G^2T (OA = 96.56%). (l) PSFormer (OA = 98.69%). We use red boxes to highlight the areas where improvements have been made.

(i.e., Single Scale/32) yields relatively higher classification performance, particularly for the SA dataset composed of homogeneous large-scale objects. The "Progressive Shrinking" strategy demonstrates improved accuracy but is constrained by the precision of object boundaries, while our "Progressive Superpixel Merging" strategy can achieve the highest performance on the Pavia Univerisity and Houston Univerisity datasets.

Table VI displays the classification accuracies on three datasets with the combination of different modules in the transformer backbone. The baseline method represents the model without a linear projection layer and using the MSA layer, the "+Linear Projection layer" represents the model that uses the linear projection layer for channel dimension mapping, and the "+Spectral MSA Layer" denotes the model that further replaces the MSA layer with the spectral MSA layer. It is observed that the linear projection layer can improve accuracies by 0.12%, 1.13%, and 2.36% on three datasets, which conduces to channelwise context mixing. Moreover, with the help of the spectral MSA layer, the accuracy is further increased by 0.16%, 0.52%, and 0.84% on three datasets, which demonstrates its effectiveness in modeling

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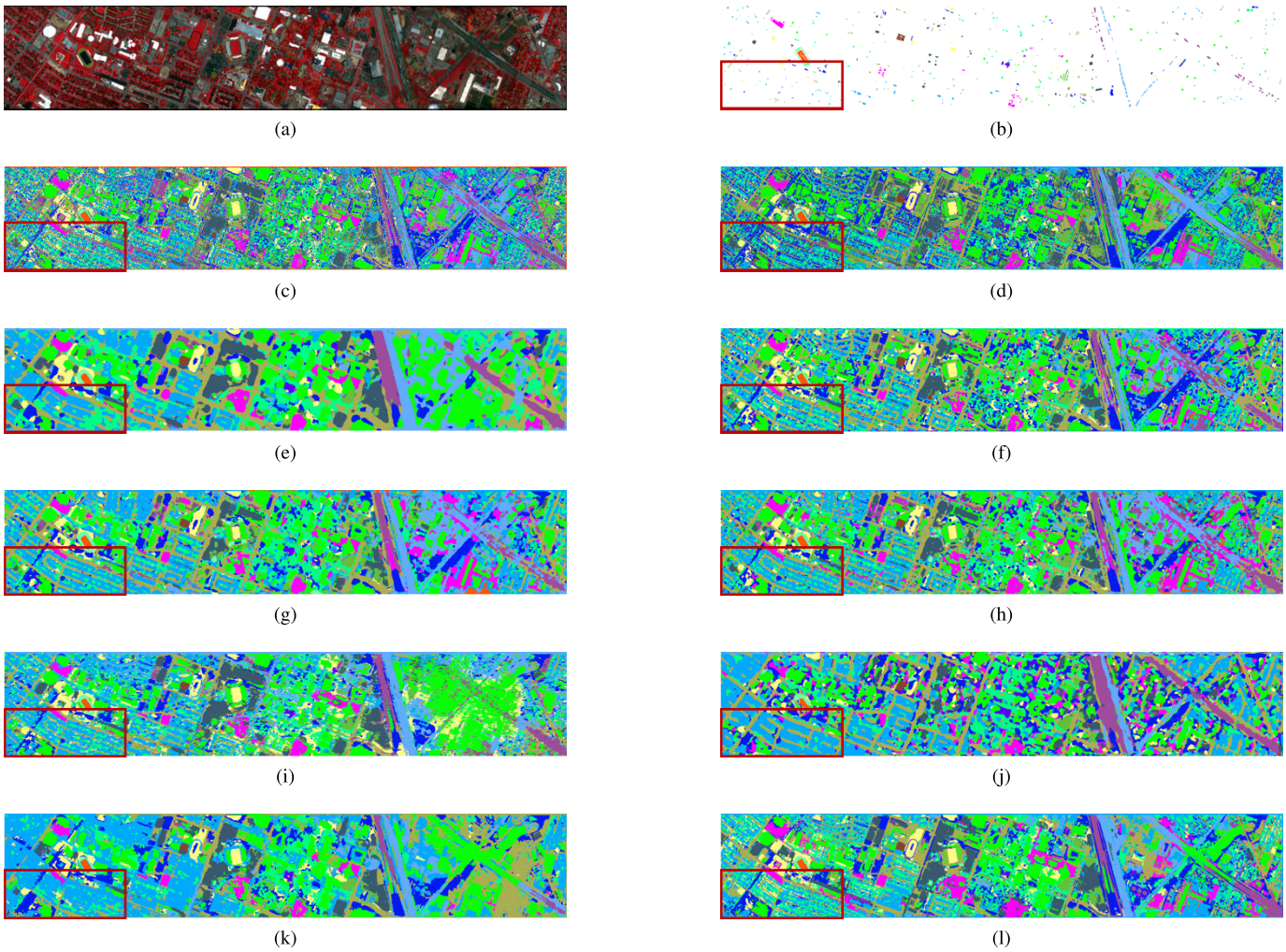


Fig. 10. Classification maps generated by different methods on the HU dataset. (a) False color image. (b) Ground-truth map. (c) SVM (OA = 85.13%). (d) SuperPCA (OA = 85.22%). (e) HPDM-SPRN (OA = 88.78%). (f) A^2S^2K-ResNet (OA = 88.87%). (g) GAHT (OA = 87.57%). (h) SSFTT (OA = 85.67%). (i) GTFN (OA = 88.00%). (j) DCN-T (OA = 71.93%). (k) G^2T (OA = 88.60%). (l) PSFormer (OA = 89.41%). We use red boxes to highlight the areas where improvements have been made.

TABLE V
OA (%) INDICES OF PSFORMER USING DIFFERENT ARCHITECTURES FOR THREE DATASETS

Method	OA		
	SA	PU	HU
Single Scale/4	85.00±1.91	87.93±2.70	64.26±3.79
Single Scale/32	**99.03±0.33**	92.03±0.93	69.12±1.43
Progressive Shrinking	97.83±1.14	92.85±2.54	77.47±2.83
Progressive Superpixel Merging	98.84±0.54	**98.69±0.30**	**89.41±1.14**

TABLE VI
OA (%) INDICES OF PSFORMER USING DIFFERENT COMBINATIONS OF MODULES IN THE TRANSFORMER BACKBONE FOR THREE DATASETS

Method	OA		
	SA	PU	HU
Baseline	98.56±0.66	97.04±1.48	86.21±0.90
+Linear Projection layer	98.68±0.71	98.17±0.84	88.57±1.19
+Spectral MSA layer	**98.84±0.54**	**98.69±0.30**	**89.41±1.14**

global relationships without substantial computational cost.

We also present the performance with four classification head designs, where the "Gate Mechanism w/o BSConv" represents the model that replaces 3 × 3 BSConv layers with standard 3 × 3 convolutional layers. As shown in Table VII, the "Add Head" and "Concat Head" both show slight advantages in different datasets. By assigning larger importance coefficients to more beneficial information for each pixel, the "Gate Mechanism w/o BSConv" effectively exploits multiscale features and improves accuracies by 0.32% and 0.58% on the SA and PU datasets. Overall, the proposed "Gate Mechanism" classification head achieves the highest accuracies on three datasets by enhancing local details and integrating multiscale feature representations adaptively.

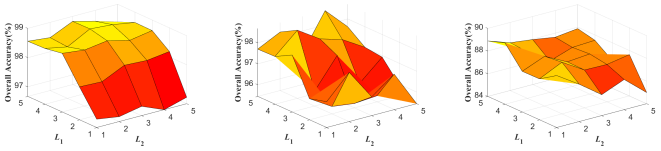
E. Parameter Sensitivity Analysis

The PSFormer method includes two sets of hyperparameters that control the network size and architecture.

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TABLE VII
OA (%) INDICES OF PSFORMER USING DIFFERENT CLASSIFICATION HEADS FOR THREE DATASETS

Method	OA		
	SA	PU	HU
Add	96.51±0.60	96.88±0.85	88.03±0.89
Concate	97.32±0.68	96.95+1.01	86.78±0.92
Gate Mechanism w/o BSConv	97.64±0.82	97.53±0.54	87.76±0.68
Gate Mechanism	**98.84±0.54**	**98.69±0.30**	**89.41±1.14**



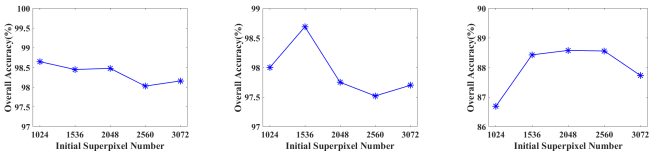
(a) (b) (c)

Fig. 11. OA (%) indices of PSFormer with different number of stages and encoder layers for three datasets. (a) SA. (b) PU. (c) HU.

1) Number of stages and encoder layers: L_1 and L_2.
2) Number of superpixels in each stage: $M^{(1)}$, $M^{(2)}$, $M^{(3)}$, and $M^{(4)}$.

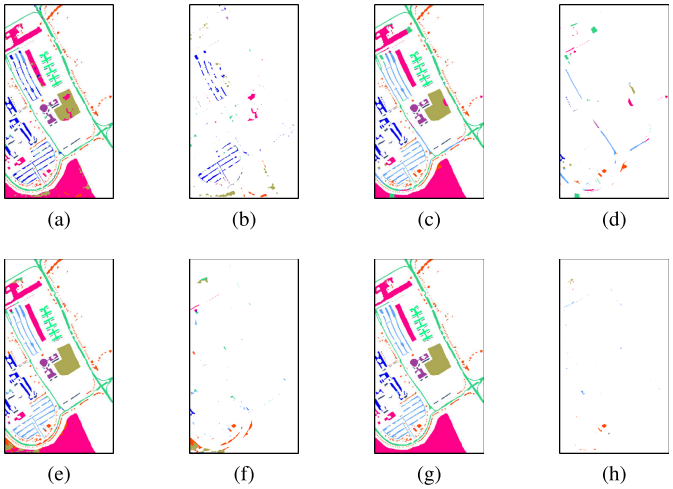
First, we analyzed the classification performance with different combinations of the number of stages and encoder layers that significantly influence the model size of PSFormer. We set the combinations of stages and encoder layers in the range of $\{1+1, 1+2, 1+3, 1+4, 1+5, 2+1, 2+2, 2+3, 2+4, 2+5, 3+1, 3+2, 3+3, 3+4, 3+5, 4+1, 4+2, 4+3, 4+4, 4+5, 5+1, 5+2, 5+3, 5+4, 5+5\}$ for three datasets. As shown in Fig. 11, PSFormer tends to yield higher performance as the number of stages increases on three datasets. Moreover, there is a relatively sharp rise in accuracy when increasing the number of stages from one to two, indicating the effectiveness of multiscale features over single-scale counterparts. However, additional stages beyond four do not necessarily lead to higher accuracy, particularly for the PU and HU datasets, which have fragmented land covers and benefit less from additional stages. Regarding the number of encoder layers, the optimal configuration varies among datasets. For the SA dataset that benefits from smoothed features to distinguish large-scale objects, more encoder layers are advantageous. In contrast, the PU and HU datasets benefit from fewer encoder layers with their fragmented land covers. To strike a balance between accuracy and efficiency, we selected the configurations "4 + 3" for the SA dataset, "4 + 2" for the PU dataset, and "4 + 1" for the HU dataset.

Second, we investigated the optimal combinations of the number of superpixels. Specifically, we set the number of superpixels in the range of $\{1024+512+256+128, 1536+768+384+192, 2048+1024+512+256, 2560+1280+640+320, 3072+1536+768+384\}$ for three datasets, where the numbers of superpixels are halved between every two stages [32]. The results are displayed in Fig. 12, it is observed that the optimal numbers of superpixels vary greatly for three datasets due to various HSI scenes and multiscale objects, which demonstrate the necessity of complementary multiscale feature extraction. Optimal superpixel numbers are determined as "1024 + 512 + 256 + 128" for SA dataset, "1536 + 768 + 384 + 192" for PU dataset, and "2048 + 1024 + 512 + 256" for HU dataset.



(a) (b) (c)

Fig. 12. OA (%) indices of PSFormer with different numbers of superpixels for three datasets. (a) SA. (b) PU. (c) HU. Note that we use the initial superpixel number $M^{(1)}$ as the label of X-axis and do not show the complete "$M^{(1)}+M^{(2)}+M^{(3)}+M^{(4)}$" for simplicity.



(a) (b) (c) (d)

(e) (f) (g) (h)

Fig. 13. Demonstration on the impact of the progressive superpixel merging strategy on the PU dataset. "Single Scale/4" strategy. (a) Classification map. (b) Difference map. "Single Scale/32" strategy. (c) Classification map. (d) Difference map. "Progressive Shrinking" strategy. (e) Classification map. (f) Difference map. "Progressive Superpixel Merging" strategy. (g) Classification map. (h) Difference map.

F. Visualization Analysis

In this section, we provide visualization results to intuitively illustrate the effectiveness of PSFormer. Fig. 13 displays the classification maps and classification difference maps obtained using different architectures (i.e., "Single Scale/4," "Single Scale/32," "Progressive Shrinking," and "Progressive Superpixel Merging") on the PU dataset. It can be seen that the models only exploiting single-scale features will suffer from severe misclassification or lose edge details, and the "Progressive Shrinking" strategy partially alleviates the problem by integrating multiscale features. Our model using the "Progressive Superpixel Merging" strategy can not only reduce misclassification but also preserve detailed information well.

In Fig. 14, the multiscale features $\mathbf{F}^{(i)}$ $(i=\{0,1,2,3,4\})$ and the corresponding normalized importance coefficients $\boldsymbol{\alpha}^{(i)}$ $(i=\{0,1,2,3,4\})$ within the gate mechanism are visualized

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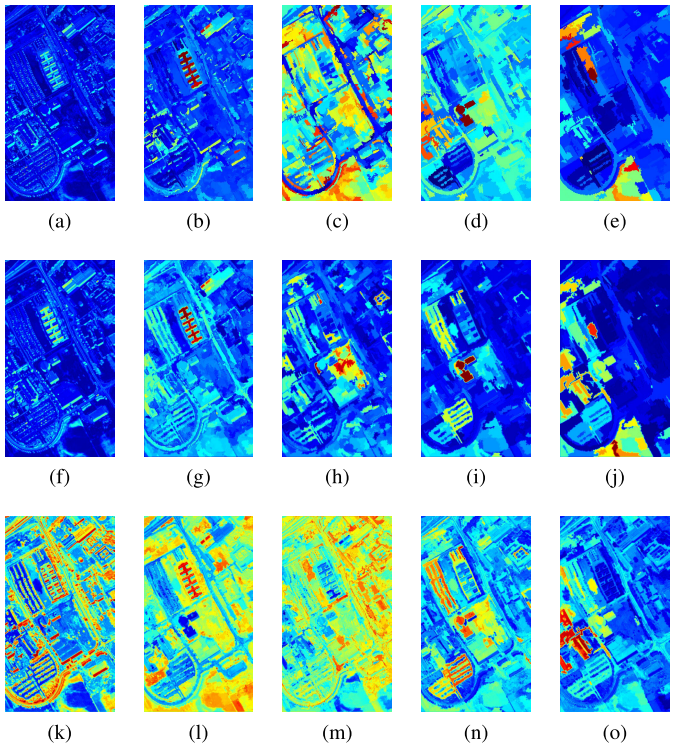


Fig. 14. Demonstration on the effect of the classification head with gate mechanism on the PU dataset. (a)–(e) Multiscale pixel feature maps before applying the BSConv layer. (f)–(j) Multiscale pixel feature maps after using the BSConv layer. (k)–(o) Normalized importance coefficient parameter of features.

for the PU dataset, where (a)–(e) presents the multiscale features before applying the BSConv layer, (f)–(j) displays the multiscale features after using the BSConv layer, and (k)–(o) shows the normalized importance coefficients of features, respectively. It is observed that multiscale feature maps contain complementary information, ranging from low-level fine details to high-level coarse semantics. Also, the BSConv layer enhances the local detail information within these feature maps. As for the normalized importance coefficients, the highly activated regions (i.e., the red parts) of low-level features can discern textures and boundaries of land covers, which is particularly beneficial for identifying small-scale objects such as "Trees" and "Shadows" [see Fig. 14(k)]. In contrast, the highly activated regions of high-level features emphasize global spatial dependencies, which can improve the integrity and reduce intraclass variance of large-scale objects distributed across various regions, including "Self-Blocking bricks" and "Gravel" [see Fig. 14(n) and (o)]. This demonstrates that the proposed classification head with gate mechanism can adaptively learn the importance coefficients and integrate features at different stages to achieve promising performance.

G. Impact of the Number of Training Samples

In real-world scenarios, it is difficult to attain huge quantities of training data without enormous resource consumption. To validate the classification performance of PSFormer when limited training data are available, we conduct classification

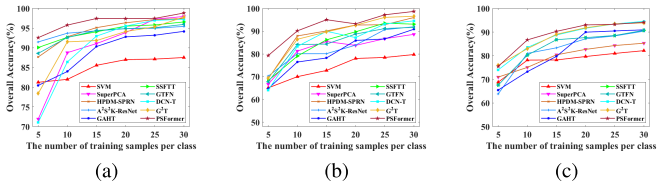


Fig. 15. OA (%) indices of PSFormer with different numbers of training samples for three datasets. (a) SA. (b) PU. (c) HU.

with different numbers of training samples in the range of {5, 10, 15, 20, 25, 30} for three datasets. As shown in Fig. 15, all methods achieve better performance as the training data amount increases. Compared with other methods, PSFormer consistently yields higher or comparative accuracies. For instance, on the SA dataset, PSFormer obtains 1.09%, 2.06%, 2.32%, 1.04%, 0.19%, and 0.88% higher accuracy than the second best method using training samples of 5, 10, 15, 20, 25, and 30, respectively. Moreover, when only using five pixels in each class for classification, we can observe a sharp decrease in accuracies for transformer-based methods (e.g., GAHT and DCN-T). As in this case, the limited training data may become a significant bottleneck for transformer models with a mass of parameters, while PSFormer alleviates this problem with the progressive superpixel merging strategy as well as spectral MSA layer and is more effective for practical applications.

H. Model Complexity and Efficiency Analysis

To analyze the complexity and efficiency of the PSFormer method, we report the trainable parameters and the number of arithmetic operations [also floating point operations (FLOPs)] of different DL-based classification methods on three datasets in Table VIII, and show the training time and test time of all classification methods on three datasets in Table IX. Note that the FLOPs of different methods are measured using the full-batch data (i.e., considering the total computational cost of 111 104, 207 400, and 664 845 patches for SA, PU, and HU datasets for the patch-based methods). As shown in Table VIII, PSFormer has relatively higher or comparable parameters compared with other patch-based transformer methods. However, PSFormer has significantly fewer parameters compared with other image-based transformer methods. In terms of the FLOPs, our PSFormer can highlight its advantages by saving redundant computation in overlapping areas and making efficient designs for the transformer model. According to Table IX, we can observe that the SSFTT has a fast inference speed, since it adopts the PCA dimensional reduction. As expected, PSFormer also requires little test time on three datasets, as it directly processes the whole HSI instead of a small patch for each batch and essentially improves efficiency by using more memory overhead.

I. Generality and Applicability of PSFormer

In order to evaluate the generality and applicability of our proposed PSFormer, we further conduct comparison experiments on two high-resolution HSI datasets collected by UAVs, including the WHU-Hi-HongHu dataset [55] representing agriculture scene and the QUH-Tangdaowan dataset [56] that

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TABLE VIII
TRAINABLE PARAMETERS AND FLOPS OF DIFFERENT DL-BASED METHODS ON THREE DATASETS

Dataset	Complexity	HPDM-SPRN	A^2S^2K-ResNet	GAHT	SSFTT	GTFN	DCN-T	G^2T	PSFormer
Salinas	Params (K)	205.40	377.00	972.62	**148.49**	169.29	11587.65	2507.94	482.76
	FLOPs (G)	18046.66	6691.88	73791.85	2710.38	18809129.47	337.31	256.78	**7.88**
Pavia University	Params (K)	184.63	220.73	927.11	**148.49**	165.85	11585.86	2493.91	334.13
	FLOPs (G)	302959.55	38447.75	187643.08	50624.27	34398119.60	630.25	476.39	**11.49**
Houston University	Params (K)	193.27	283.35	1369.36	**148.49**	169.23	11587.39	2500.01	205.63
	FLOPs (G)	1018356.38	961552.04	4173165.58	162282.02	112510389.66	2019.15	1531.23	**40.18**

TABLE IX
RUNNING TIME OF DIFFERENT CLASSIFICATION METHODS ON THREE DATASETS

Dataset	Running Time	SVM	SuperPCA	HPDM-SPRN	A^2S^2K-ResNet	GAHT	SSFTT	GTFN	DCN-T	G^2T	PSFormer
Salinas	Train. (s)	**2.2**	6.1	75.9	286.6	139.5	24.2	914.8	5759.8	149.9	43.8
	Test. (s)	3.7	7.3	23.5	40.2	49.2	4.9	394.9	22.7	37.4	**1.3**
Pavia University	Train. (s)	**1.2**	3.9	44.0	450.4	50.5	11.7	551.8	8329.9	457.8	90.6
	Test. (s)	3.3	7.1	25.0	71.0	38.5	5.9	311	29.6	41.3	**2.3**
Houston University	Train. (s)	**45.8**	90.4	419.7	2393.1	137.9	52.0	3603.1	20790.4	256.1	177.5
	Test. (s)	50.2	197.6	88.0	20.8	177.5	**1.5**	56.7	32.5	183.0	6.8

TABLE X
ACCURACY COMPARISONS ON THE WHU-HI-LONGKOU DATASET WITH 100 TRAINING SAMPLES AND THE QUH-TANGDAOWAN DATASET WITH 30 TRAINING SAMPLES

Dataset	Indices	SVM	SuperPCA	HPDM-SPRN	A^2S^2K-ResNet	GAHT	SSFTT	GTFN	DCN-T	G^2T	PSFormer
WHU-Hi-HongHu	OA	74.58±0.41	82.51±0.37	95.90±0.51	92.92±1.14	95.33±0.67	94.07±0.89	93.79±0.72	95.36±0.44	96.48±0.17	**96.59±0.54**
	AA	70.90±0.18	84.10±0.31	95.80±0.66	94.05±0.45	95.63±0.44	94.51±0.28	95.42±0.53	96.59±0.27	**96.62±0.20**	96.26±0.45
	Kappa	69.11±0.46	78.56±0.39	94.82±0.64	91.16±1.38	94.12±0.84	92.57±1.09	92.23±0.86	94.17±0.54	95.57±0.21	**95.69±0.68**
QUH-Tangdaowan	OA	71.88±2.40	77.94±1.82	85.13±1.45	77.13±1.38	76.02±2.92	72.23±3.44	77.54±0.94	73.56±1.59	90.02±1.50	**92.50±1.05**
	AA	82.78±0.84	87.18±1.01	92.35±0.43	87.42±0.64	85.97±0.71	83.59±0.55	86.15±0.83	84.28±0.92	93.22±0.35	**95.08±0.53**
	Kappa	69.27±2.48	75.72±1.92	83.46±1.55	74.88±1.41	73.61±3.02	69.63±3.49	75.15±1.02	70.83±1.75	88.79±1.61	**91.55±1.16**

represents complex wetland scene. Following the experimental setup in [55], 100 samples are randomly selected from each class as the training set, and the rest are used as the test set to evaluate performance on the WHU-Hi-HongHu dataset. Also, 30 training samples are randomly chosen from each class for the QUH-Tangdaowan dataset. The parameters of the comparison methods are carefully tuned to obtain optimal performance. Regarding the PSFormer, we set the numbers of stages and encoder layers to "4 + 3" and "4 + 5" for the WHU-Hi-HongHu and QUH-Tangdaowan datasets, respectively. The numbers of superpixels are set to 2048 for the WHU-Hi-HongHu dataset and 1024 for the QUH-Tangdaowan dataset. The classification results are presented in Table X. As can be seen, PSFormer consistently achieves the best classification results in terms of OA and Kappa metrics among all comparison methods. On the WHU-Hi-HongHu dataset, the accuracies of PSFormer are 0.11% and 0.12% higher than that of the second-best method. Meanwhile, the accuracies of PSFormer are 2.48% and 2.76% higher than that of the second-best method for the QUH-Tangdaowan dataset. This demonstrates that PSFormer not only has superior classification performance but also shows better robustness across various HSI scenarios. Additional results including individual class accuracies and classification maps will be completely provided at https://github.com/immortal13.

V. CONCLUSION

In this article, a novel PSFormer (i.e., PSFormer) was proposed for HSI classification, which is not tied to the patch-based learning framework and can take full advantage of the transformer to exploit multiscale local–global feature representations. To capture beneficial contextual information while remaining computationally efficient, we introduce the progressive superpixel merging strategy, which can control the scale of feature maps with simple matrix multiplication. Furthermore, the spectral MSA layer is devised to model

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nonlocal interactions from the spectral perspective, which can enhance the global consistency of classification results with linear complexity. In addition, we employ the classification head with gate mechanism to adaptively combine feature representations at different stages for classification. Extensive experimental results on three benchmark HSI datasets and two UAV HSI datasets illustrate the effectiveness of the proposed PSFormer method.

Nevertheless, this work has some limitations. For instance, PSFormer trades space for time and requires a relatively high-maximum GPU memory. In addition, the empirical predefinition of superpixel segmentation scales might impact accuracies, replacing it with alternative learnable superpixel segmentation techniques could potentially enhance performance. In the future, we will concentrate on investigating efficient multiscale ViT architectures for large-scale remote sensing image processing and applications.

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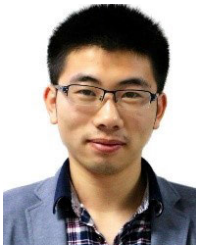
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